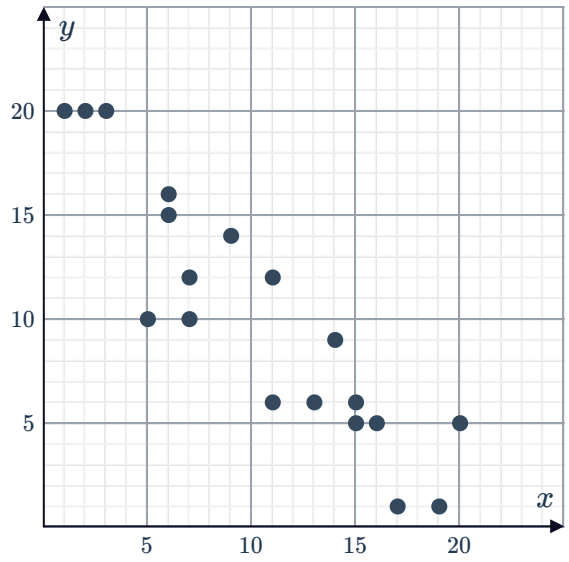
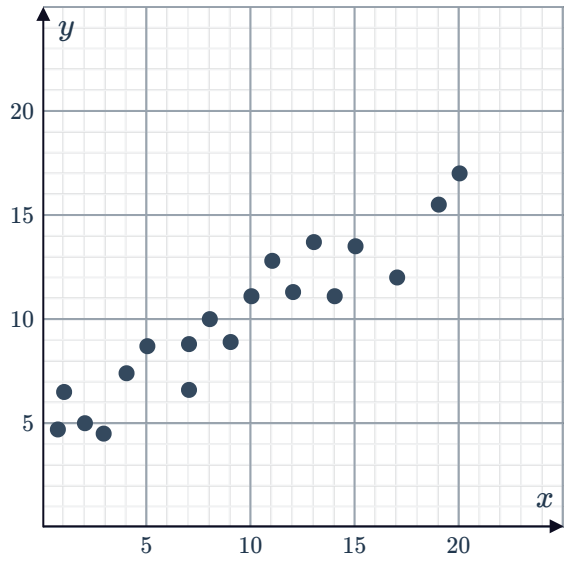
Line of best fit

1 Draw an approximate line of best fit by hand for each of the the scatter plots below:

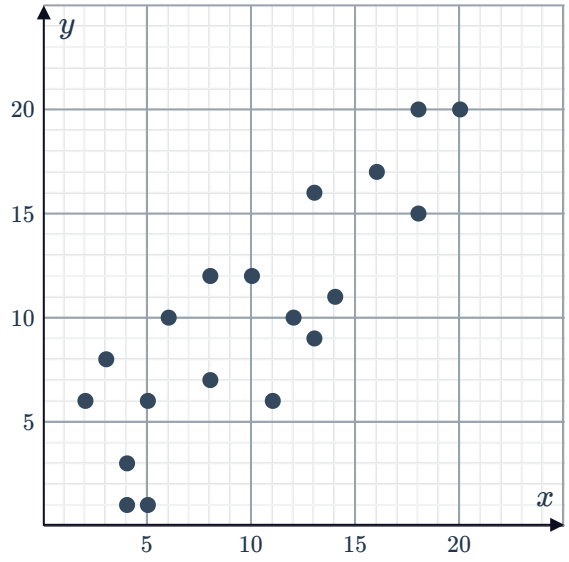
a



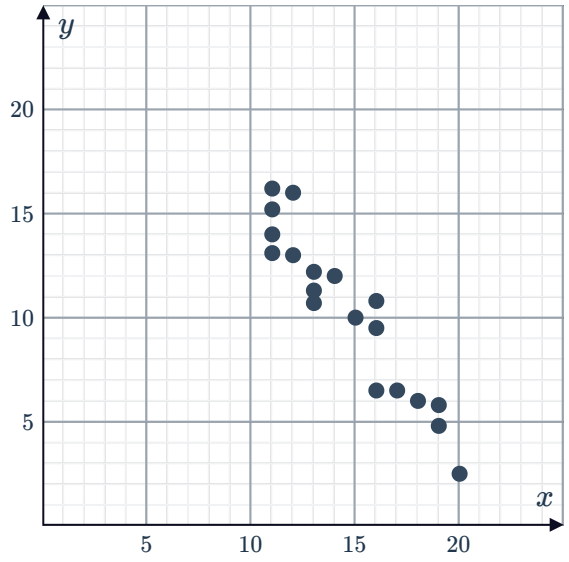
b



c



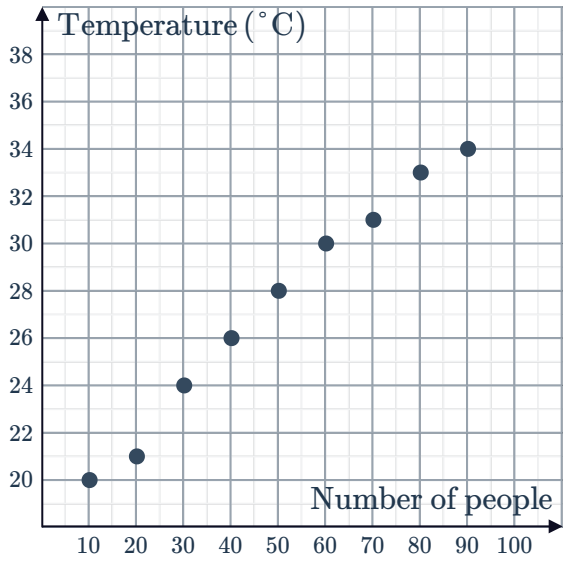
d



2 The following scatter plot displays data for the number of people in a room and the room temperature, as collected by a researcher:

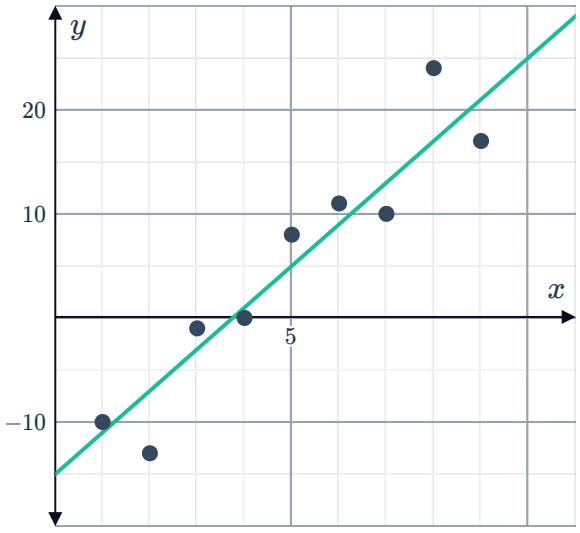
a Find the mean point.

b Draw a line of best fit through this mean point.



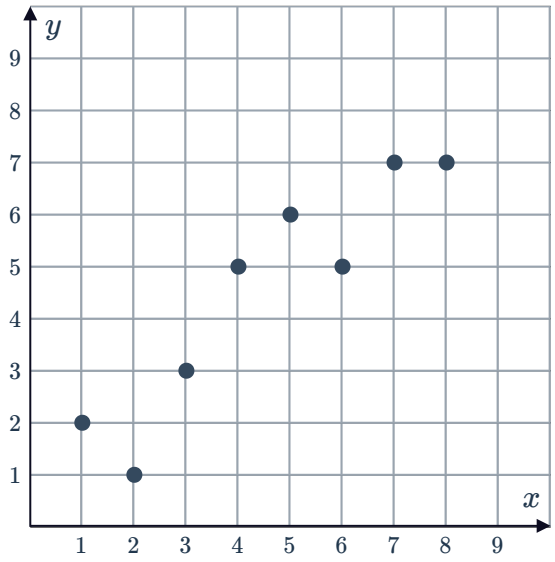
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3 Find the equation of the line of best fit for the data shown:



4 The equation for the line of best fit is given by $P = -4t + 116$, where t is time. Determine if the relationship shows that over time, P is increasing, decreasing or remain constant.

5 The following scatter plot shows the data for two variables, x and y:



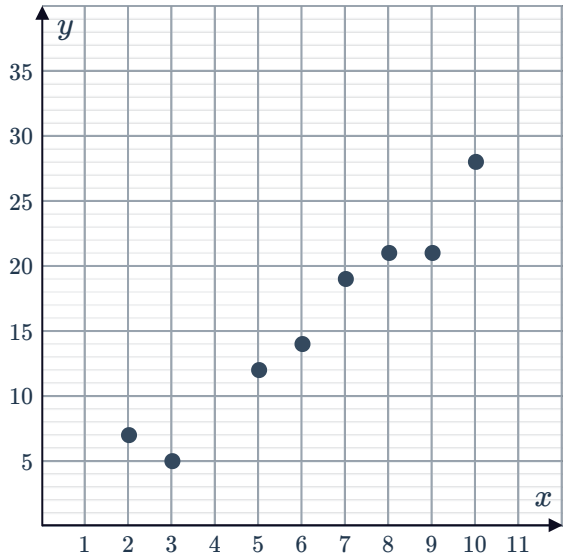
a Sketch the line of best fit for this data.

b Use your line of best fit to estimate the value of y when:

i $x = 4.5$ **ii** $x = 9$

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6 Consider the data plotted on the following scatter plot:

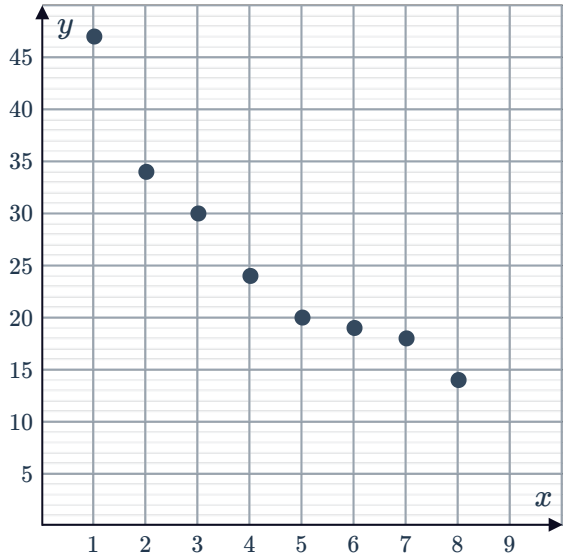


a Is there a negative or positive relationship between the x and y variables?

b Sketch the line of best fit for this data.

c Find the equation of your line of best fit.

7 Consider the data plotted on the following scatter plot:



a Is there a negative or positive relationship between the x and y variables?

b Sketch the line of best fit for this data.

c Find the equation of your line of best fit.

8 A least squares regression line is given by $y = 3.59x + 6.72$.

a State the gradient of the line.

b Does the gradient of the line indicate that the bivariate data set has a positive or negative correlation?

c As x increases by 1 unit, how does the value of y change?

d State the value of the y-intercept.

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9 A least squares regression line is given by $y = -3.67x + 8.42$.

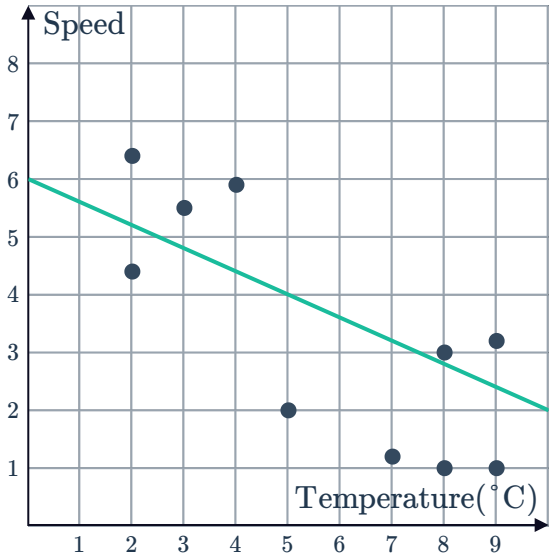
- **a** State the gradient of the line.
- **b** Does the gradient of the line indicate that the bivariate data set has a positive or negative correlation?
- **c** As x increases by 1 unit, how does the value of y change?
- **d** State the value of the y-intercept.

10 A least squares regression line is given by $T = -7.26S + 4.35$.

- **a** State the gradient of the line.
- **b** Does the gradient of the line indicate that the bivariate data set has a positive or negative correlation?
- **c** As x increases by 1 unit, how does the value of y change?
- **d** State the value of the vertical intercept.

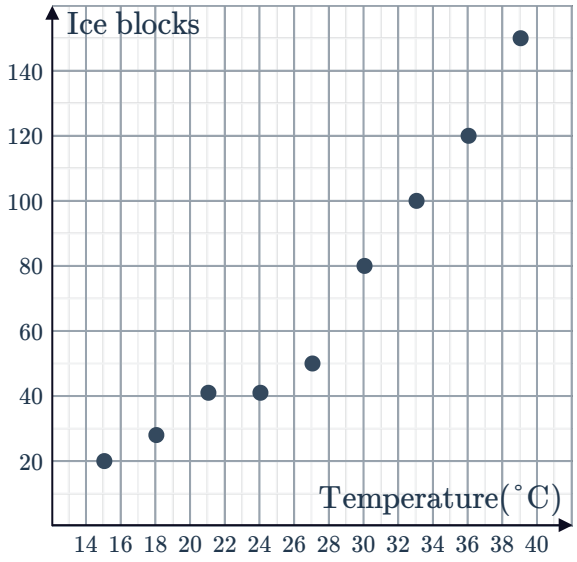
Applications

11 The average monthly temperature and the average wind speed, in knots, in a particular location was plotted over several months. The graph shows the points for each month's data and their line of best fit:
Use the line of best fit to approximate the wind speed on a day when the temperature is 5°C.



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12 The following scatter plot graphs data for the number of ice blocks sold at a shop on days with different temperatures.



a Sketch the line of best fit for this data.

b Use your answer line of best fit to estimate the number of ice blocks that will be sold on a:

i $31^\circ C$ day **ii** $42^\circ C$ day

c Does the number of ice blocks sold increase or decrease as the temperature increases?

13 Jordano measures his heart rate at various times while running. The data is shown below in the table:

Time (minutes)	0	2	4	6	8	10
Heart Rate (BPM)	58	54	58	67	67	74

a Plot the data and sketch the line of best fit.

b Which data point is closest to the line of best fit?

c Which data point is furthest from the line of best fit?

14 A dam used to supply water to the neighboring town had the following data recorded for its volume over a number of months:

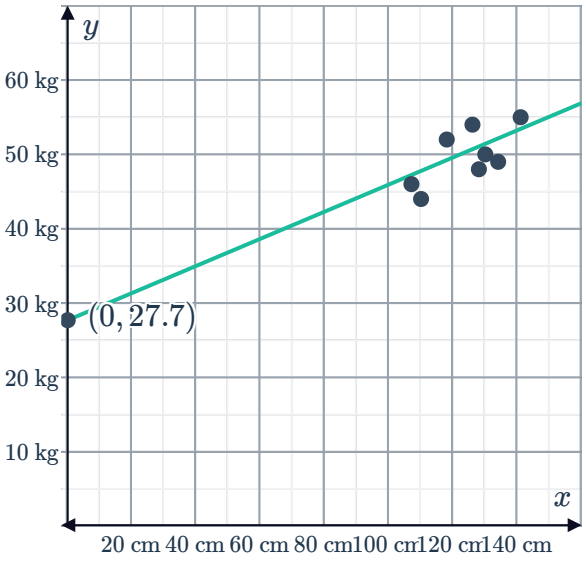
Month	1	2	3	4
Volume (billion of litres)	116	106	104	92

a Plot the data on a number plane.

b Sketch the line of best fit for this data.

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15 The heights (in cm) and the weights (in kg) of 8 primary school children is shown on the scattergraph below:



- **a** State the y-value of the y-intercept.
- **b** Describe the meaning of the y-intercept in context.
- **c** Does the interpretation in the previous part make sense in this context? Explain your answer.

16 The amount of money households spend on dining out each week, D, is measured against their weekly income, I.
The following linear regression model is fitted to the data: $D = 0.3I + 27$.

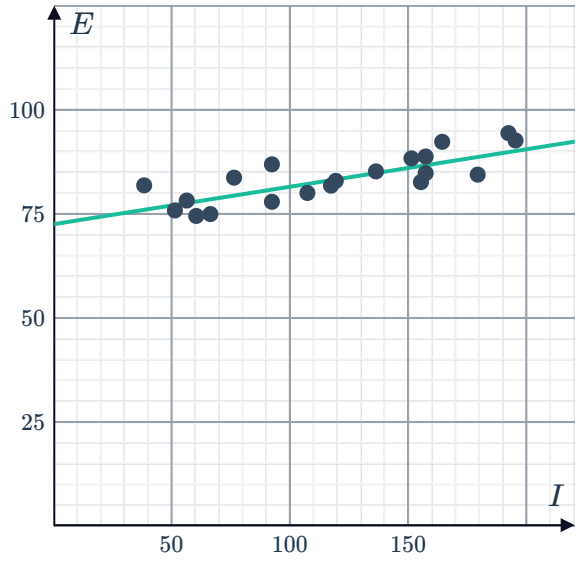
- **a** Explain the meaning of the vertical intercept.
- **b** Does the interpretation in the previous part make sense in this context? Explain your answer.
- **c** State the gradient of the line.
- **d** If the weekly income of a family increases by $\$200$, by how much can we expect their spending on dining out to increase?

17 The number of hours spent watching TV each evening, h, is measured against the percentage results, m, achieved in the Economics exam.
The following linear regression model is fitted to the data: $m = -10h + 97$.

- **a** Explain the meaning of the vertical intercept.
- **b** Does the interpretation in the previous part make sense in this context? Explain your answer.
- **c** State the gradient of the line.
- **d** If a student increases the amount of TV they watch by 3.5 hours, by how much can we expect their Economic exam mark to drop?

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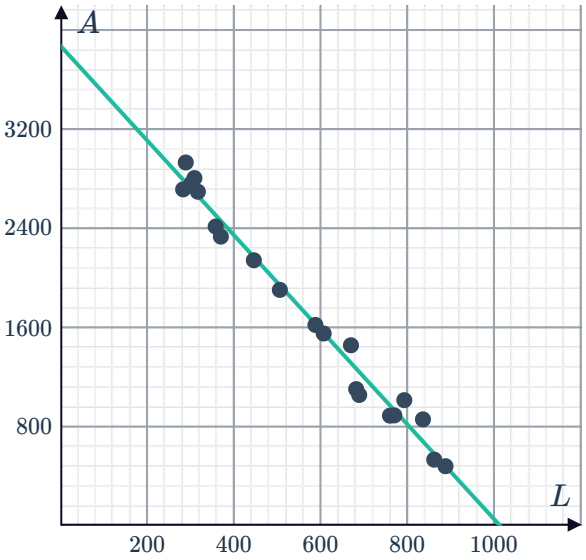
18 The life expectancy (E), in years, of individuals at different annual incomes (I), per $\$1000$, is shown:
The equation of the line of best fit is $E = 0.09I + 72.55$.



a By how much does average life expectancy change for each $\$1000$ of annual income?

b Find the average life expectancy of someone who earns no income.

19 Scientists record the number of aphids (A) in areas with different numbers of ladybeetles (L), and graph the data in the scatterplot shown:
The equation of the line of best fit is $A = -3.82L + 3865.21$.



a How much does the average aphid population change by with each extra ladybeetle? Give your answer to the nearest aphid.

b Find the average aphid population of a region with no ladybeetles. Give your answer to the nearest aphid.

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